



GC1015

HYPOTHESIS TESTS, EFFECTS, PRACTICAL SIGNIFICANCE, AND A/B GUARDRAILS

Quantitative Methods for Business Analysis · Session 6



Use what you already know

RECALL

Decision Frame and Running-Case Brief

1

Recall

- Why must an operational decision track a guardrail metric alongside the primary target key performance indicator (**KPI**)?

2

Connect

- In Unit **S06.U05**, we synthesize statistical uplift with operational guardrail metrics to make defensible feature launch decisions.

Not in your PNP: follow along; no written response here.

CHOOSE, THEN EXPLAIN

Why must an operational decision track a guardrail metric alongside the primary target KPI?

- A** A guardrail reveals harm hidden by an improving target KPI.
- B** Meeting the target KPI means other outcomes are acceptable.
- C** A guardrail replaces the need to measure the target KPI.

Choose first. Compare your reason with a neighbor.

Not in your PNP: follow along; no written response here.



Use what you already know

RECALL

Operational KPI Definitions and Statistical Meaning

1

Recall

- When subtracting an 85.0 percent target from an observed 78.4 percent rate, how is the direct difference expressed?

2

Connect

- In Units **S06.U03** and **S06.U04**, two-proportion tests and confidence intervals measure absolute percentage-point uplift between test variants.

Not in your PNP: follow along; no written response here.

CHOOSE, THEN EXPLAIN

When subtracting an 85.0 percent target from an observed 78.4 percent rate, how is the direct difference expressed?

- ☐ A The difference is negative 6.6 percentage points.
- ☐ B The difference is negative 6.6 percent.
- ☐ C The difference is positive 6.6 percentage points.

Choose first. Compare your reason with a neighbor.



Three steps toward a
launch recommendation

TODAY'S INVESTIGATION

Three steps toward a launch recommendation

1

Key point

- 1. **Hypothesis Testing Logic, Errors, and Benchmark Mean Comparisons:** Test a claim against a benchmark.
- 2. **A/B Testing with Proportions and Statistical Uplift:** Estimate and test a conversion difference.

2

Key point

- 3. **Practical Significance, Guardrail Metrics, and Cross-Tool Reconciliation:** Check value, harm, and reproducibility.

Not in your PNP: follow along; no written response here.



Slides hand off to the lab,
the lab hands back

TODAY'S LOOP: SLIDES AND LAB

Every section ends in your notebook and workbook; every practice returns you here

1

Sections before the break

- After each section's retrieval slide, a hand-off slide sends you to the lab for one 10 to 10 minute practice: Excel first, Statistical Package for the Social Sciences (**SPSS**) second, with a slide-case example and a fresh example.

2

Break after Section 2

- Return for Sections 3, then the guided build. Complete the planned applied work.

3

Then the studio and the close

- Group studio (**A/B** Experimentation Launch Evaluation and Guardrail Dilemma), synthesis and exit ticket. Open now: the lab notebook in Colab and the workbook in Excel.

Not in your PNP: follow along; no written response here.

WHY IT MATTERS

WHY THIS SESSION MATTERS

A purchase increase can be statistically detectable yet still fall short of a business goal or create costly customer harm. You will learn to report what the sample supports, check secondary risks, and make a recommendation with clear limits.



Not in your PNP: follow along; no written response here.



Case briefing

CEDARCART'S CHECKOUT DECISION

Case briefing

1

Key point

- The fictional mobile commerce app CedarCart's product team serves eligible mobile app shoppers through a checkout flow. It is comparing the standard checkout, A, with a streamlined version, B, before deciding on nationwide deployment.
- During the experiment window, each variant has 400 randomly assigned users. Purchases and complaints are counted per assigned user.

2

Key point

- The team requires at least 8 percentage points of purchase uplift and no more than a 1-point complaint increase. Your output is a bounded launch recommendation.

Not in your PNP: follow along; no written response here.

DECISION POINT: CEDARCART

CedarCart product team



Situation

CedarCart observed 160 purchases and 8 complaints among 400 users assigned to A, versus 200 purchases and 32 complaints among 400 assigned to B. The product team must decide whether to deploy B nationwide.

A

Deploy B now

Benefit: More purchases were observed with B.

Risk: Customer complaints may make deployment costly.

B

Hold B for revision

Benefit: The team can investigate the complaint increase.

Risk: It delays a possible purchase gain.

Your decision

Would you deploy variant B nationwide now or hold it for revision, and which evidence should decide? Make a provisional choice; revisit it at the end of class.

Not in your PNP: follow along; no written response here.

SECTION

HYPOTHESIS TESTING LOGIC, ERRORS, AND BENCHMARK MEAN COMPARISONS

Formulate directional and non-directional statistical hypotheses, balance Type I and II decision errors, and evaluate benchmark goal attainment using a one-sample t-test. Formulating Hypotheses and Error Trade-offs · One-Sample t-Test for Benchmark Goals

Not in your PNP: follow along; no written response here.



Formulating Hypotheses
and Error Trade-offs

WHAT STATISTICAL HYPOTHESES MEAN IN BUSINESS DECISIONS

Formulating Hypotheses and Error Trade-offs

1

Meaning

- **Parameter Versus Statistic:** A parameter describes an unobserved population, such as the true purchase probability p . A sample statistic describes observed data from sample shoppers.
- **Formulating H_0 and H_1 :** The null hypothesis **H_0** claims no true operational difference between variant A and variant B. The alternative hypothesis **H_1** claims a genuine difference exists.

2

Meaning

- **CedarCart Checkout Guardrail:** CedarCart compares standard checkout A against streamlined checkout B. Formulating hypotheses in advance prevents teams from adjusting targets after inspecting sample noise, ensuring disciplined decision rules.

In your PNP: complete the purple blanks.

TWO-TAILED HYPOTHESIS FORMULATION AND TYPE I ERROR PROBABILITY (1/4)

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Symbols:

α : The significance level, defining the maximum acceptable probability of a Type I error (false positive) under the null model.

H_0 : The null hypothesis specifying no effect or no difference in the underlying population parameter.

$P(\cdot \mid \cdot)$: Conditional probability of taking a rejection action given that the null condition actually holds.

Not in your PNP: follow along; no written response here.

TWO-TAILED HYPOTHESIS FORMULATION AND TYPE I ERROR PROBABILITY (2/4)

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Inputs: Management specifies the maximum acceptable risk of a false positive rollout, conventionally set to $\alpha = 0.05$ or $\alpha = 0.01$ prior to experiment launch.

Steps:

1. State the null hypothesis H_0 : $p_B - p_A = 0$ and alternative hypothesis H_1 : $p_B - p_A \neq 0$.
2. Choose alpha based on the business severity of committing a false positive deployment.
3. Define critical rejection regions in the tails of the test statistic distribution corresponding to area alpha.

Not in your PNP: follow along; no written response here.

TWO-TAILED HYPOTHESIS FORMULATION AND TYPE I ERROR PROBABILITY (3/4)

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Steps:

4. Reject H_0 if and only if the test statistic falls into the critical region, yielding $p\text{-value} \leq \alpha$.

Result: A decision boundary that bounds the long-run false positive rate to exactly α under repeated sampling from an unchanged population.

Alpha represents an insurance policy against deploying ineffective operational changes. Setting $\alpha = 0.05$ means that if variant B is identical to variant A, there is only a 5% chance of mistakenly adopting variant B.

Not in your PNP: follow along; no written response here.

TWO-TAILED HYPOTHESIS FORMULATION AND TYPE I ERROR PROBABILITY (4/4)

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Use: Used across all confirmatory A/B experimentation platforms and quality control programs before allocating capital to operational rollouts.

Limit: Controlling alpha does not protect against Type II errors (beta) or systemic sampling bias. A test with alpha = 0.05 may still have very low power to detect subtle improvements.

Not in your PNP: follow along; no written response here.

THE MECHANISM OF TYPE I AND TYPE II DECISION ERRORS

Formulating Hypotheses and Error Trade-offs

Managerial Action	H0 is True (No Difference)	H0 is False (B Differs from A)
Reject H0 (Deploy B)	Type I Error (False Positive, prob = alpha): Waste rollout capital on equal screen	Correct Decision (True Positive, prob = 1 - beta): Successfully capture genuine conversion uplift
Fail to Reject H0 (Retain A)	Correct Decision (True Negative, prob = 1 - alpha): Avoid unnecessary deployment cost	Type II Error (False Negative, prob = beta): Missed opportunity to improve conversions

The decision matrix defines the four possible intersections of managerial action and underlying population truth.

All future eligible CedarCart mobile app shoppers | Fall 2026 pre-rollout evaluation | Unit: managerial consequence

Not in your PNP: follow along; no written response here.



Formulating Hypotheses
and Error Trade-offs

THE MECHANISM OF TYPE I AND TYPE II DECISION ERRORS

Formulating Hypotheses and Error Trade-offs

1

How it works

- **Two Realities and Two Errors:** Reality either satisfies **H0** or **H1**. A Type I error rejects a true null hypothesis.
- A Type **II** error fails to reject a false null hypothesis.

2

How it works

- **Error Probabilities Alpha and Beta:** Alpha measures false-positive risk under **H0**. Beta measures false-negative risk when a true operational effect exists. Statistical power equals 1 minus beta.
- **Managerial Risk Balance:** For CedarCart, alpha risks deploying an ineffective checkout screen at high capital cost. Beta risks shelving a genuinely superior checkout experience.

In your PNP: complete the purple blanks.



One-Sample t-Test for
Benchmark Goals

THE MATHEMATICAL MECHANISM OF THE ONE-SAMPLE T-STATISTIC

One-Sample t-Test for Benchmark Goals

1

Standard Error Scaling

- When the population standard deviation is unknown, the standard error is calculated as sample standard deviation divided by the square root of sample size, reflecting sampling variability across repeated samples.

2

Student t-Distribution and

- Degrees of Freedom
- Estimating variability from data adds uncertainty, requiring the heavier-tailed t-distribution indexed by degrees of freedom equal to n minus 1.

3

The Standardized Ratio

- The t-statistic measures the exact number of standard errors separating the observed sample mean from the operational benchmark target.

In your PNP: complete the purple blanks.

ONE-SAMPLE T-TEST STATISTIC AND CONFIDENCE INTERVAL (1/3)

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}, \quad df = n - 1, \quad CI_{1-\alpha} = \bar{x} \pm t_{\alpha/2, df} \cdot \left(\frac{s}{\sqrt{n}} \right)$$

Symbols:

t: Test statistic measuring standardized distance between sample mean and target

$\bar{x}$: Observed sample mean

μ_0 : Prespecified operational benchmark target

s: Sample standard deviation

n: Number of independent observations in the sample

df: Degrees of freedom equal to sample size minus one

CI_(1- α): Confidence interval for the true population mean

$t_{(\alpha/2, df)}$: Two-tailed critical value from Student t-distribution

Inputs: $\bar{x} = 72.0$, $\mu_0 = 70.0$, $s = 8.0$, $n = 16$, $\alpha = 0.05$

Not in your PNP: follow along; no written response here.

ONE-SAMPLE T-TEST STATISTIC AND CONFIDENCE INTERVAL (2/3)

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}, \quad df = n - 1, \quad CI_{1-\alpha} = \bar{x} \pm t_{\alpha/2, df} \cdot \left(\frac{s}{\sqrt{n}} \right)$$

Steps:

1. $df = 16 - 1 = 15$
2. $SE = 8.0 / \sqrt{16} = 8.0 / 4.0 = 2.0$
3. $t = (72.0 - 70.0) / 2.0 = 2.0 / 2.0 = 1.00$
4. $p = \text{Excel: } =T.DIST.2T(1.00, 15) \approx 0.3329$
5. $t_{\text{crit}} = \text{Excel: } =T.INV.2T(0.05, 15) \approx 2.131$
6. Margin of Error = $2.131 \times 2.0 = 4.262$
7. 95% CI = $72.0 \pm 4.262 = [67.74, 76.26]$

Not in your PNP: follow along; no written response here.

ONE-SAMPLE T-TEST STATISTIC AND CONFIDENCE INTERVAL (3/3)

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}, \quad \text{df} = n - 1, \quad \text{CI}_{1-\alpha} = \bar{x} \pm t_{\alpha/2, \text{df}} \cdot \left(\frac{s}{\sqrt{n}} \right)$$

Result: $t = 1.00$, $\text{df} = 15$, $p = 0.333$, 95% CI = [67.74, 76.26] points

At significance level $\alpha = 0.05$, we fail to reject the null hypothesis. The sample data are consistent with a true population mean of 70 points.

Use: Used when auditing operational continuous performance indicators against internal standards or contractual service level agreements.

Limit: Assumes independent sampling, interval-scaled data, and near-normal distribution in small samples; cannot fix non-response or selection bias.

Not in your PNP: follow along; no written response here.

CEDARCART SATISFACTION SURVEY WORKED EVIDENCE

One-Sample t-Test for Benchmark Goals

Metric	Formula / Source	Value	Unit
Sample Size (n)	Survey count	16	users
Sample Mean ($\bar{x}$)	Survey average	72	points
Sample Std Dev (s)	Survey dispersion	8	points
Benchmark Target (μ_0)	Management standard	70	points
Standard Error (SE)	$s / \sqrt{n}$	2	points
Test Statistic (t)	$(\bar{x} - \mu_0) / SE$	1	ratio
Two-Sided p-value	T.DIST.2T(1.00, 15)	0.333	probability

The sample mean of 72 points exceeds the 70-point benchmark, but yields $t = 1.00$ and $p = 0.333$, failing to demonstrate a statistically significant population difference.

Variant B survey respondents | Seven-day test window | Unit: Satisfaction index points (0-100 scale)

Not in your PNP: follow along; no written response here.



One-Sample t-Test for
Benchmark Goals

CEDARCART SATISFACTION SURVEY WORKED EVIDENCE

One-Sample t-Test for Benchmark Goals

1

Step-by-Step Calculation Degrees of freedom equal 15. The standard error is $8.0 / \sqrt{16} = 2.0$ points.

- In a post-checkout survey of $n = 16$ variant B users, mean satisfaction was 72.0 points with standard deviation 8.0 points against a 70.0-point target.

2

Worked example: CedarCart Survey Inputs

- The test statistic is $t = (72.0 - 70.0) / 2.0 = 1.00$. ### Statistical Evaluation Excel function **T.DIST.2T**(1.00, 15) yields $p = 0.333$.
- The 95 percent confidence interval of [67.74, 76.26] points includes the 70-point benchmark.

Not in your PNP: follow along; no written response here.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

1

Key point

- Suppose customer satisfaction survey responses were voluntary and only 16 of 200 contacted users replied. Why would a statistically significant t-test still fail to support a product launch?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

Not in your PNP: follow the activity directions here.

DECISION POINT: HYPOTHESIS TESTING LOGIC, ERRORS, AND BENCHMARK MEAN COMPARISONS

Use the evidence, explain your decision, and name one limit.



Key Point

CedarCart observes a survey satisfaction mean of 72 points against a 70-point operational benchmark with $p = 0.333$. What is the valid conclusion?

In your PNP: use the matching slide page.



Excel first, SPSS second

LAB PRACTICE 1: BENCHMARK T-TEST FOR CUSTOMER SATISFACTION

10 minutes in the lab; then slide 28.

1

Open and do

- **Colab:** heading Class practice 1: test the satisfaction benchmark. Excel cells: **B7, B8, C7, D7**.
- **Excel:** Calculate standard error in **B9**, t-statistic in **B10**, and two-tailed p-value in **B11** using. **SPSS:** Run One-Sample T Test on satisfaction variable against test value.

2

Check and bring back

- Calculated t-statistic of 1.00 and p-value of 0.333 confirming failure to reject the 70-point benchmark. Timer 10 min, debrief 2 min, then return to **A/B** Testing with Proportions and Statistical Uplift (slide 28).

3

Help or go further

- **If you need help:** Open Toolkit Section 1.1, inspect the benchmark formula, and calculate standard error for sample size 16.
- **If you finish early:** Critique this benchmark test: how would an edge condition of zero sample variance affect the calculated t-statistic?

In your PNP: use this page for the lab handoff.

SECTION

A/B TESTING WITH PROPORTIONS AND STATISTICAL UPLIFT

Execute and interpret a pooled two-proportion z-test and construct an unpooled confidence interval to quantify experimental conversion uplift. Randomized Units and Two-Proportion z-Test Mechanics · Interval Estimation for Absolute Difference

Not in your PNP: follow along; no written response here.

RANDOMIZED UNITS AND CONVERSION PROPORTIONS

Randomized Units and Two-Proportion z-Test Mechanics

Variant	Assigned Users	Purchasers	Conversion Rate
Variant A (Standard)	400	160	0.4
Variant B (Shorter)	400	200	0.5
Total Combined	800	360	0.45

Variant B achieves a 50.0% conversion rate versus 40.0% for Variant A across 400 assigned users per arm.

Eligible U.S. CedarCart mobile app shoppers | Seven-day post-assignment window | Unit: Users and conversion proportions

Not in your PNP: follow along; no written response here.



Randomized Units and
Two-Proportion z-Test
Mechanics

RANDOMIZED UNITS AND CONVERSION PROPORTIONS

Randomized Units and Two-Proportion z-Test Mechanics

1

Meaning: Core Definition

- The User as Assignment Unit An experiment assigns eligible participants randomly across variants.
- The denominator must count unique assigned users, not sessions or clicks, to satisfy statistical independence and avoid survivorship bias.

2

CedarCart Baseline CedarCart evaluates 800 randomized users

- Variant A standard checkout has 160 purchases from 400 users (40.0%). Variant B shorter checkout has 200 purchases from 400 users (50.0%), representing a 10 percentage point absolute gain.

In your PNP: complete the purple blanks.

POOLED TWO-PROPORTION Z-TEST STATISTIC (1/3)

$$z = \frac{(\hat{p}_B - \hat{p}_A) - 0}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}, \quad \hat{p}_0 = \frac{x_A + x_B}{n_A + n_B}$$

Symbols:

$\hat{p}_A, \hat{p}_B$: Sample conversion proportions in Variant A and Variant B

x_A, x_B : Count of successful conversions in Variant A and Variant B

n_A, n_B : Total assigned eligible users in Variant A and Variant B

$\hat{p}_0$: Pooled conversion proportion under the null hypothesis H_0

z : Standard normal test statistic

Inputs: Variant A: $x_A = 160, n_A = 400$. Variant B: $x_B = 200, n_B = 400$.

Steps:

1. Calculate observed sample rates: $\hat{p}_A = 160 / 400 = 0.40, \hat{p}_B = 200 / 400 = 0.50$.

Not in your PNP: follow along; no written response here.

POOLED TWO-PROPORTION Z-TEST STATISTIC (2/3)

$$z = \frac{(\hat{p}_B - \hat{p}_A) - 0}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}, \quad \hat{p}_0 = \frac{x_A + x_B}{n_A + n_B}$$

Steps:

2. Compute pooled rate: $p_hato = (160 + 200) / (400 + 400) = 360 / 800 = 0.45$.
3. Compute pooled variance factor: $0.45 * (1 - 0.45) = 0.45 * 0.55 = 0.2475$.
4. Compute reciprocal sample size sum: $1/400 + 1/400 = 0.0025 + 0.0025 = 0.005$.
5. Calculate null standard error: $SE0 = \text{sqrt}(0.2475 * 0.005) = \text{sqrt}(0.0012375) = 0.035178$.
6. Compute standardized test statistic: $z = (0.50 - 0.40) / 0.035178 = 0.10 / 0.035178 = 2.843$.

Not in your PNP: follow along; no written response here.

POOLED TWO-PROPORTION Z-TEST STATISTIC (3/3)

$$z = \frac{(\hat{p}_B - \hat{p}_A) - 0}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}, \quad \hat{p}_0 = \frac{x_A + x_B}{n_A + n_B}$$

Steps:

7. Compute two-tailed p-value using standard normal distribution: $p = 2 * (1 - \text{Phi}(2.843)) = 0.00447$.

Result: $z = 2.843$, $p = 0.0045$

The observed 10 percentage point uplift sits 2.843 standard errors above zero, rejecting the null hypothesis at $\alpha = 0.05$.

Use: Used to test whether conversion proportions differ between two randomized variants in A/B experiments.

Limit: Assumes independent user assignments and sample sizes large enough that successes and failures exceed 10 in each arm.

Not in your PNP: follow along; no written response here.

EXCEL TWO-PROPORTION Z-TEST AND SPSS VERIFICATION (1/2)

```
=LET(  
    nA, 400, xA, 160,  
    nB, 400, xB, 200,  
    pA, xA / nA,  
    pB, xB / nB,  
    p0, (xA + xB) / (nA + nB),  
    se0, SQRT(p0 * (1 - p0) * (1/nA + 1/nB)),
```

Symbols:

nA, nB: Sample sizes for control and treatment groups xA, xB: Success counts in control and treatment groups pA, pB: Observed group conversion proportions

p0: Combined pooled proportion under the null model se0: Standard error of the difference under H0 z: Computed test statistic pval: Two-tailed significance value

Inputs: Cells containing nA=400, xA=160, nB=400, xB=200.

Steps: 1. Assign cell inputs to named variables inside LET formula. 2. Calculate group conversion rates $pA = 0.40$ and $pB = 0.50$. 3. Calculate pooled parameter $p0 = 0.45$.

Not in your PNP: follow along; no written response here.

EXCEL TWO-PROPORTION Z-TEST AND SPSS VERIFICATION (2/2)

```
z, (pB - pA) / se0,  
pval, 2 * (1 - NORM.S.DIST(ABS(z), TRUE)),  
HSTACK(z, pval)  
)
```

Steps: 4. Evaluate standard error $se0 = 0.035178$. 5. Compute test statistic $z = 2.84268$. 6. Compute two-tailed probability $pval = 0.00447$. 7. In SPSS, run CROSSTABS /TABLES=group BY purchase /STATISTIC=CHISQ to verify Pearson chi-square = 8.081.

Result: $z = 2.843$, $p\text{-value} = 0.0045$

Reconciles spreadsheet calculations with enterprise statistical packages without manual rounding discrepancy.

Use: Applied in automated business analytics dashboards and verification sheets for A/B testing pipelines.

Limit: Excel formula assumes non-empty numeric inputs; missing cells must be screened before evaluating LET arrays.

Not in your PNP: follow along; no written response here.



Randomized Units and
Two-Proportion z-Test
Mechanics

THE MECHANISM OF THE POOLED NULL STANDARD ERROR

Randomized Units and Two-Proportion z-Test Mechanics

1

How it works

- Null Hypothesis Model. The null hypothesis **H₀** specifies that Variant A and Variant B share an identical population conversion parameter p .
- Under this assumption of equality, we pool purchase counts across both arms to construct the minimum-variance parameter estimate. - Null Standard Error Calculation.

2

How it works

- The pooled proportion is 360 divided by 800, which equals 0.45 (45.0 percent).
- Null standard error **SE₀** equals the square root of 0.45 times 0.55 times the sum of reciprocal sample sizes ($1/400 + 1/400$), yielding 0.03518, or 3.518 percentage points.

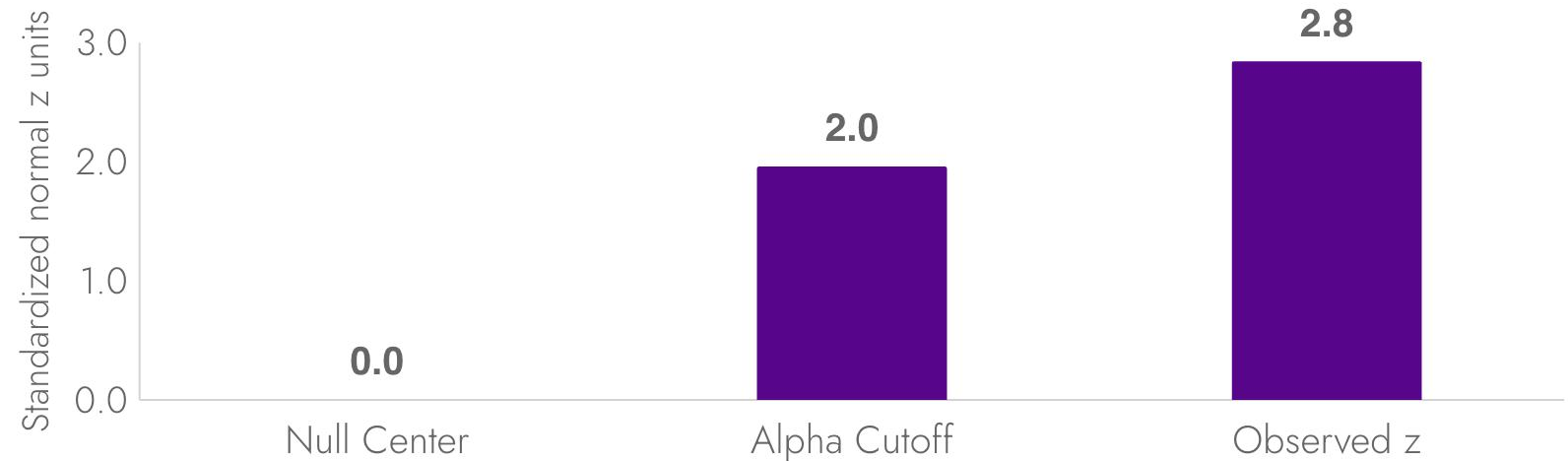
In your PNP: complete the purple blanks.



Randomized Units and
Two-Proportion z-Test
Mechanics

WORKED EVIDENCE: CALCULATING THE Z-STATISTIC AND P-VALUE

Randomized Units and Two-Proportion z-Test Mechanics



The observed test statistic $z = 2.843$ lies well beyond the two-tailed 5% alpha critical cutoff of 1.960.

Hypothetical null distribution sampling units | Experiment benchmark | Unit: Standardized normal z units

Not in your PNP: follow along; no written response here.



Randomized Units and
Two-Proportion z-Test
Mechanics

WORKED EVIDENCE: CALCULATING THE Z-STATISTIC AND P-VALUE

Randomized Units and Two-Proportion z-Test Mechanics

1

Statistical Decision Critical cutoff for $\alpha = 0.05$ is 1.960. The two-tailed p-value is 0.0045, well below 0.05. We reject the null hypothesis of equal conversion rates.

- $0.50 - 0.40 = 0.10$ (10 percentage points). Standardizing against **SEO**: $z = 0.10 / 0.035178 = 2.843$. Null pooling assumes both variants share an identical population rate.

In your PNP: complete the purple blanks.



Interval Estimation for
Absolute Difference

MATHEMATICAL MECHANISM OF THE UNPOOLED STANDARD ERROR

Interval Estimation for Absolute Difference

1

How it works

- Why Use Unpooled Variance? Hypothesis testing assumes equal parameters under **H₀**, justifying pooled counts. Interval estimation measures the plausible difference between two distinct distributions without assuming equality.
- Sum of Variances Principle. For independent groups, the variance of the difference equals the sum of individual group variances.

2

How it works

- The unpooled standard error equals the square root of $p_{\text{hat_A}} \times (1 - p_{\text{hat_A}}) / n_{\text{A}}$ plus $p_{\text{hat_B}} \times (1 - p_{\text{hat_B}}) / n_{\text{B}}$.

Not in your PNP: follow along; no written response here.

CHOOSE, THEN EXPLAIN

Why Use Unpooled Variance?

- ☐ A The interval estimates a difference without assuming equal group rates.
- ☐ B The interval assumes both groups share one rate.
- ☐ C The interval ignores uncertainty within each group.

Choose first. Compare your reason with a neighbor.

Not in your PNP: follow along; no written response here.

UNPOOLED CONFIDENCE INTERVAL FOR DIFFERENCE BETWEEN TWO PROPORTIONS (1/3)

$$CI_{95\%} = (\hat{p}_B - \hat{p}_A) \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}}$$

Symbols:

$\hat{p}_A, \hat{p}_B$: Observed sample conversion proportions in arms A and B
 n_A, n_B : Total randomized eligible users assigned to arms A and B

$z_{(1-\alpha/2)}$: Standard normal critical multiplier (1.96 for a 95% confidence level)
 $CI_{(95\%)}$: Plausible interval range for the true population difference ($p_B - p_A$)

Inputs: $n_A = 400, x_A = 160$ ($\hat{p}_A = 0.40$); $n_B = 400, x_B = 200$ ($\hat{p}_B = 0.50$); confidence level = 0.95 ($z = 1.96$).

Steps: 1. Calculate observed difference: $0.50 - 0.40 = 0.10$ (10.0 percentage points)

Not in your PNP: follow along; no written response here.

UNPOOLED CONFIDENCE INTERVAL FOR DIFFERENCE BETWEEN TWO PROPORTIONS (2/3)

$$CI_{95\%} = (\hat{p}_B - \hat{p}_A) \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}}$$

Steps:

2. Compute variance of arm A: $(0.40 * 0.60) / 400 = 0.24 / 400 = 0.000600$
3. Compute variance of arm B: $(0.50 * 0.50) / 400 = 0.25 / 400 = 0.000625$
4. Sum variances and take square root: $\text{sqrt}(0.000600 + 0.000625) = \text{sqrt}(0.001225) = 0.0350$
5. Multiply unpooled standard error by critical z: $1.96 * 0.0350 = 0.0686$
6. Construct bounds: $0.10 - 0.0686 = 0.0314$ and $0.10 + 0.0686 = 0.1686$

Not in your PNP: follow along; no written response here.

UNPOOLED CONFIDENCE INTERVAL FOR DIFFERENCE BETWEEN TWO PROPORTIONS (3/3)

$$CI_{95\%} = (\hat{p}_B - \hat{p}_A) \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}}$$

Result: 95% CI = [0.0314, 0.1686] (or 3.14 to 16.86 percentage points).

We are 95 percent confident that the true population increase in checkout conversion for Variant B over Variant A lies between 3.14 and 16.86 percentage points.

Use: Quantifying parameter uncertainty and effect size in randomized A/B experimentation and conversion rate optimization.

Limit: Requires independent randomized units and adequate sample sizes (at least 10 successes and 10 failures per group). Does not correct for non-random attrition or sample ratio mismatch.

Not in your PNP: follow along; no written response here.

EXCEL DYNAMIC CALCULATION OF UNPOOLED CONFIDENCE INTERVAL (1/3)

```
=(p_B - p_A) - NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))  
=(p_B - p_A) + NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))
```

Symbols:

NORM.S.INV(0.975): Generates standard normal critical value $z = 1.95996$ (rounds to 1.96)

SQRT (square root of the expression): Computes the unpooled standard error across independent sample proportions

p_A, p_B : Cell references containing observed conversion rates for arms A and B

n_A, n_B : Cell references containing assigned sample sizes for arms A and B

Not in your PNP: follow along; no written response here.

EXCEL DYNAMIC CALCULATION OF UNPOOLED CONFIDENCE INTERVAL (2/3)

```
=(p_B - p_A) - NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))  
=(p_B - p_A) + NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))
```

Inputs: Cell B2 (n_A) = 400, Cell C2 (purchases_A) = 160, Cell D2 (p_A) = 0.40; Cell B3 (n_B) = 400, Cell C3 (purchases_B) = 200, Cell D3 (p_B) = 0.50.

Steps:

1. In Cell D4, enter formula for difference: =D3-D2 (returns 0.1000)
2. In Cell D5, enter unpooled SE formula: =SQRT((D2*(1-D2)/B2)+(D3*(1-D3)/B3)) (returns 0.0350)
3. In Cell D6, enter lower bound: =D4-NORM.S.INV(0.975)*D5 (returns 0.0314)

Not in your PNP: follow along; no written response here.

EXCEL DYNAMIC CALCULATION OF UNPOOLED CONFIDENCE INTERVAL (3/3)

```
=(p_B - p_A) - NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))  
=(p_B - p_A) + NORM.S.INV(0.975) * SQRT((p_A * (1 - p_A) / n_A) + (p_B * (1 - p_B) / n_B))
```

Steps:

4. In Cell D7, enter upper bound: =D4+NORM.S.INV(0.975)*D5 (returns 0.1686)

Result: Lower bound = 0.0314 (3.14%), Upper bound = 0.1686 (16.86%).

Implements an audit-compliant, dynamic spreadsheet calculation for unpooled confidence intervals directly from summary counts.

Use: A/B testing analysis templates and executive reporting dashboards in Microsoft Excel.

Limit: Assumes input cells contain valid non-zero sample sizes and evaluated proportions between 0 and 1. Fails if empty cells are evaluated as text.

Not in your PNP: follow along; no written response here.

WORKED EVIDENCE FOR CEDARCART A/B UPLIFT

Interval Estimation for Absolute Difference

Variant	Assigned Users	Purchasers	Conversion Rate	Lower 95% Bound	Upper 95% Bound
A: Standard	400	160	40.0%	-	-
B: Shorter	400	200	50.0%	-	-
Difference (B - A)	-	-	+10.0 pp	+3.14 pp	+16.86 pp

Variant B increases purchase conversion by 10.0 percentage points, with an unpooled 95 percent confidence interval from 3.14 to 16.86 percentage points.

Eligible U.S. mobile app users | Seven-day observation window | Unit: Users and conversion percentages

Not in your PNP: follow along; no written response here.



Interval Estimation for
Absolute Difference

WORKED EVIDENCE FOR CEDARCART A/B UPLIFT

Interval Estimation for Absolute Difference

1

CedarCart Uplift Calculation

- **Group A:** 160 purchases / 400 users = 40.0% conversion.
- **Group B:** 200 purchases / 400 users = 50.0% conversion (difference: +10.0 pp, +25.0% relative).

2

Margin and Interval

- Unpooled **SE** = $\sqrt{(0.40 * 0.60 / 400) + (0.50 * 0.50 / 400)} = 0.0350$.
- **95% Confidence Interval:** $0.1000 \pm 1.96 * 0.0350 = [3.14, 16.86]$ percentage points.

3

Estimation Limit

- Valid inference requires keeping all 800 randomized users in the denominator without attrition.

Not in your PNP: follow along; no written response here.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

1

Key point

- In CedarCart's complaint analysis, the 95 percent confidence interval for the complaint rate increase was [3.01, 8.99] percentage points against a 1.0 point guardrail.
- What does this indicate for the product launch?

2

Partner check

- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

Not in your PNP: follow the activity directions here.

DECISION POINT: A/B TESTING WITH PROPORTIONS AND STATISTICAL UPLIFT

Use the evidence, explain your decision, and name one limit.



Key Point

Variant B converts at 50 percent versus 40 percent for Variant A. Why is reporting this result as a 10 percent lift misleading to leadership?

In your PNP: use the matching slide page.



Excel first, SPSS second

PRACTICE 2: TEST CONVERSION AND ESTIMATE UPLIFT

10 minutes of practice; 2-minute debrief; then the break

1

Open and do

- **Colab:** heading Class practice 2: test conversion and estimate its effect. Excel cells: **B7, B8, C7, D7**.
- Calculate pooled rate, standard error, z and p in **B9:B12**; estimate the 95% uplift interval in **B14:B15**. Run **SPSS H2**: group by purchase, **CHISQ**.

2

Check and bring back

- Bring back $z = 2.843$ and the uplift interval $[3.14, 16.86]$ percentage points. Save Checkpoint_B; return for the debrief, then the break (slide 52).

3

Help or go further

- **Need help:** Read Part B's supplied formula notes and compare pooled test error with unpooled interval error.
- **Finished early:** Explain how unequal group sizes would affect the pooled standard error.

In your PNP: use this page for the lab handoff.



Pause. Reset. Return ready.



PAUSE BEFORE THE LAUNCH AUDIT

Return for Lab Part C

Stretch, Hydrate, and Reset

Not in your PNP: follow along; no written response here.

SECTION

PRACTICAL SIGNIFICANCE, GUARDRAIL METRICS, AND CROSS-TOOL RECONCILIATION

Synthesize statistical test outputs with practical uplift hurdles, secondary guardrail metrics, and cross-tool software verification to make bounded launch decisions. Practical Significance and A/B Guardrails · Tool Reconciliation and Reproducibility in Excel and SPSS

Not in your PNP: follow along; no written response here.



Practical Significance and
A/B Guardrails

WHAT PRACTICAL SIGNIFICANCE AND GUARDRAILS MEAN

Practical Significance and A/B Guardrails

1

Meaning: Statistical Versus Practical

- Significance Statistical significance filters out sampling noise under the null hypothesis. Practical significance determines whether an effect size justifies commercial deployment and ongoing operating costs.

2

Protecting Operations with

- Guardrails A guardrail metric sets a strict boundary on secondary operational risks, such as customer complaints or page latency.
- Breaching a prespecified guardrail halts deployment even when conversion uplift is statistically detectable.

In your PNP: complete the purple blanks.

NET INCREMENTAL ECONOMIC IMPACT OF SCREEN VARIANT B (1/3)

$$\Delta\text{Profit} = (\Delta\text{Purchases} \times m) - (N_B \times c_{\text{rollout}}) - (\Delta\text{Complaints} \times c_{\text{complaint}})$$

Symbols:

Δ Purchases: Incremental purchase count in Variant B relative to baseline rate

m: Unit contribution margin per incremental purchase

N_B: Total exposed users assigned to Variant B

c_{rollout}: Implementation and infrastructure cost per exposed user

Δ Complaints: Incremental support complaints generated by Variant B

c_(complaint): Unit operational resolution cost per filed complaint

Not in your PNP: follow along; no written response here.

NET INCREMENTAL ECONOMIC IMPACT OF SCREEN VARIANT B (2/3)

$$\Delta\text{Profit} = (\Delta\text{Purchases} \times m) - (N_B \times c_{\text{rollout}}) - (\Delta\text{Complaints} \times c_{\text{complaint}})$$

Inputs: Baseline purchase rate = 0.40; Variant B purchases = 200 of 400 users (Delta Purchases = 200 - 160 = 40); $m = 12$; $N_B = 400$; $c_{\text{rollout}} = 0.60$; Baseline complaints = 8; Variant B complaints = 32 (Delta Complaints = 24); $c_{\text{complaint}} = \$25$.

Steps:

1. Calculate gross incremental margin gain: 40 purchases * \$12/purchase = \$480.
2. Calculate fixed rollout operating cost: 400 users * \$0.60/user = \$240.
3. Calculate direct customer service complaint burden: 24 incremental complaints * \$25/complaint = \$600.

Not in your PNP: follow along; no written response here.

NET INCREMENTAL ECONOMIC IMPACT OF SCREEN VARIANT B (3/3)

$$\Delta\text{Profit} = (\Delta\text{Purchases} \times m) - (N_B \times c_{\text{rollout}}) - (\Delta\text{Complaints} \times c_{\text{complaint}})$$

Steps:

4. Compute net economic impact: $\$480 - \$240 - \$600 = -\360 .

Result: Net incremental impact = $-\$360$ per 400 assigned users

Although Variant B generates 40 additional purchases, the direct customer service handling costs and operating expenses exceed the gross margin generated by $\$360$.

Use: CedarCart uses this equation to translate conversion uplift and side effect metrics into net operational value before committing engineering resources.

Limit: This calculation assumes static unit costs and excludes long-term churn, customer lifetime value erosion, and brand reputational damage.

Not in your PNP: follow along; no written response here.



Practical Significance and
A/B Guardrails

THE MECHANISM LINKING PRACTICAL THRESHOLDS AND GUARDRAILS

Practical Significance and A/B Guardrails

1

How it works: The Pre-Experiment Hurdle

- Framework Management defines four operational hurdles before observing outcomes: the test error rate alpha, the statistical test, the minimum practical uplift target, and the maximum allowable harm guardrail.

2

Multi-Dimensional Decision Criteria

- Confidence intervals must be evaluated against the practical uplift threshold.
- Concurrently, secondary guardrail metrics are audited to ensure that core performance does not create destructive side effects or net operating losses.

In your PNP: complete the purple blanks.

CEDARCART EXPERIMENTATION EVIDENCE AND UNIT ECONOMICS

Practical Significance and A/B Guardrails

Metric	Variant A (Standard)	Variant B (Shorter)	Difference (B - A)	Decision Rule
Assigned Users	400	400	0	Balanced 1:1 ratio
Purchasing Users	160	200	40	Target: $\geq +8.0$ points
Purchase Conversion Rate	40.0%	50.0%	+10.0 points	Statistically significant (p=0.0045)
Users with Complaints	8	32	24	Guardrail: $\leq +1.0$ point
Complaint Rate	2.0%	8.0%	+6.0 points	Guardrail VIOLATED (+6.0 vs +1.0)

Variant B shows higher purchase conversion but an unacceptable spike in customer complaints that creates negative net economic returns.

Assigned eligible app users | Seven-day observation window | Unit: Users and Percentages

Not in your PNP: follow along; no written response here.



Practical Significance and
A/B Guardrails

CEDARCART EXPERIMENTATION EVIDENCE AND UNIT ECONOMICS

Practical Significance and A/B Guardrails

1

Worked example: CedarCart Experimentation Metrics

- **Variant A:** 160 purchases / 400 users (40.0%); 8 complaints (2.0%)
- **Variant B:** 200 purchases / 400 users (50.0%); 32 complaints (8.0%)

2

Worked example: CedarCart Experimentation Metrics

- **Uplift:** 10.0 percentage points, 95% **CI** [3.14, 16.86] points

Not in your PNP: follow along; no written response here.



Practical Significance and
A/B Guardrails

CEDARCART EXPERIMENTATION EVIDENCE AND UNIT ECONOMICS | CONTINUED 2

Practical Significance and A/B Guardrails

1

Net Unit Economics Calculation

- **Gross margin gain:** 40 purchases * \$12 = \$480
- **Rollout cost:** 400 users * \$0.60 = -\$240

2

Net Unit Economics Calculation

- **Complaint cost:** 24 complaints * \$25 = -\$600
- **Net economic impact:** \$480 - \$240 - \$600 = -\$360, violating guardrails and profitability.

Not in your PNP: follow along; no written response here.

WORKED CROSS-TOOL RECONCILIATION ON CEDARCART DATA

Tool Reconciliation and Reproducibility in Excel and SPSS

Analysis Step	Excel Metric Value	SPSS Output Label	SPSS Metric Value
Arm A Conversion	160 / 400 (40.0%)	Group 0 Row %	40.0%
Arm B Conversion	200 / 400 (50.0%)	Group 1 Row %	50.0%
Difference (B - A)	+10.0 percentage points	Observed Uplift	+10.0 percentage points
Test Statistic	$z = 2.8427$ ($z^2 = 8.081$)	Pearson Chi-Square	8.081
Degrees of Freedom	df = 1	df	1
Two-Sided P-Value	p = 0.0045	Asymptotic Sig. (2-sided)	0.0045

The uncorrected Pearson Chi-Square from SPSS exactly matches the squared z-statistic from Excel, verifying numerical reconciliation.

Eligible CedarCart mobile app users | 7-day randomized experiment | Unit: Assigned users and test metrics

Not in your PNP: follow along; no written response here.



Tool Reconciliation and
Reproducibility in Excel
and SPSS

WORKED CROSS-TOOL RECONCILIATION ON CEDARCART DATA

Tool Reconciliation and Reproducibility in Excel and SPSS

1

Excel Calculation The pooled standard error is **0.035178**, producing a z-statistic of **2.8427**. Squaring this value yields **8.0808**.

- Arm A yielded 160 purchases from 400 users (40.0 percent). Arm B yielded 200 purchases from 400 users (50.0 percent). The pooled rate is 45.0 percent.

2

SPSS Cross-Tabulation Output SPSS reports an uncorrected **Pearson Chi-Square of 8.081 with 1 degree of**

- freedom and an asymptotic two-sided significance of 0.0045, confirming exact software reconciliation across analytical platforms.

Not in your PNP: follow along; no written response here.

SPSS SYNTAX FOR TWO-BY-TWO CROSS-TABULATION AND CHI-SQUARE VALIDATION (1/4)

```
CROSSTABS
```

```
  /TABLES=group BY purchase  
  /STATISTIC=CHISQ  
  /CELLS=COUNT ROW EXPECTED.
```

```
CROSSTABS
```

```
  /TABLES=group BY complaint  
  /STATISTIC=CHISQ  
  /CELLS=COUNT ROW EXPECTED.
```

Symbols:

group: Binary independent variable coded 0 for Arm A (standard) and 1 for Arm B (shorter checkout)

purchase: Binary primary KPI indicator coded 1 for purchased within 7 days, 0 otherwise

complaint: Binary guardrail indicator coded 1 for filed complaint within 7 days, 0 otherwise

/STATISTIC=CHISQ: Instructs SPSS to compute Pearson Chi-Square, Continuity Correction, and Fisher exact statistics

/CELLS=COUNT ROW EXPECTED: Prints observed counts, percentage within group, and expected frequencies under independence

Not in your PNP: follow along; no written response here.

SPSS SYNTAX FOR TWO-BY-TWO CROSS-TABULATION AND CHI-SQUARE VALIDATION (2/4)

```
CROSSTABS
```

```
  /TABLES=group BY purchase  
  /STATISTIC=CHISQ  
  /CELLS=COUNT ROW EXPECTED.
```

```
CROSSTABS
```

```
  /TABLES=group BY complaint  
  /STATISTIC=CHISQ  
  /CELLS=COUNT ROW EXPECTED.
```

Inputs: An active, validated dataset containing exactly 800 individual user rows without case weights or duplicate identifiers.

Steps:

1. Execute the first CROSSTABS command to generate the group-by-purchase contingency table.
2. Verify that the total row counts equal 400 for group 0 and 400 for group 1, with 0 missing cases.
3. Read the Pearson Chi-Square row: locate the test value 8.081, $df = 1$, and Asymptotic Sig. (2-sided) = 0.0045.

Not in your PNP: follow along; no written response here.

SPSS SYNTAX FOR TWO-BY-TWO CROSS-TABULATION AND CHI-SQUARE VALIDATION (3/4)

CROSSTABS

```
/TABLES=group BY purchase  
/STATISTIC=CHISQ  
/CELLS=COUNT ROW EXPECTED.
```

CROSSTABS

```
/TABLES=group BY complaint  
/STATISTIC=CHISQ  
/CELLS=COUNT ROW EXPECTED.
```

Steps:

4. Execute the second CROSSTABS command to generate the group-by-complaint contingency table.
5. Verify that complaint counts equal 8 in group 0 (2.0%) and 32 in group 1 (8.0%).
6. Read the Pearson Chi-Square row for complaints: locate test value 15.360, $df = 1$, and Asymptotic Sig. < 0.001 .

Result: The purchase table yields an uncorrected Pearson Chi-Square of 8.081 ($p = 0.0045$). The complaint table yields a Chi-Square of 15.360 ($p < 0.001$).

Not in your PNP: follow along; no written response here.

SPSS SYNTAX FOR TWO-BY-TWO CROSS-TABULATION AND CHI-SQUARE VALIDATION (4/4)

CROSSTABS

```
/TABLES=group BY purchase  
/STATISTIC=CHISQ  
/CELLS=COUNT ROW EXPECTED.
```

CROSSTABS

```
/TABLES=group BY complaint  
/STATISTIC=CHISQ  
/CELLS=COUNT ROW EXPECTED.
```

The primary conversion increase from 40.0% to 50.0% is statistically detectable at $\alpha = 0.05$. However, the secondary complaint rate increase from 2.0% to 8.0% is also statistically significant, establishing a severe guardrail violation.

Use: Use this standardized SPSS syntax in weekly experimentation reviews to cross-validate spreadsheet calculations and verify that secondary risk metrics are formally evaluated.

Limit: SPSS syntax assumes clean record-level data; it cannot detect whether duplicate pings or biased dropouts corrupted the raw database prior to file ingestion.

Not in your PNP: follow along; no written response here.

EQUIVALENCE FORMULA: TWO-PROPORTION SQUARED Z-STATISTIC TO PEARSON CHI-SQUARE (1/4)

$$z^2 = \left(\frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}} \right)^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \chi^2$$

Symbols:

$\hat{p}_A, \hat{p}_B$: Observed sample conversion proportions in Arm A and Arm B

$\hat{p}_0$: Pooled sample proportion under the null hypothesis, calculated as $(x_A + x_B) / (n_A + n_B)$

n_A, n_B : Total assigned sample sizes for Arm A and Arm B

O_{ij} : Observed count in row i , column j of the two-by-two contingency table

E_{ij} : Expected count in row i , column j under independence, calculated as $(\text{Row Total} * \text{Column Total}) / \text{Grand Total}$

Not in your PNP: follow along; no written response here.

EQUIVALENCE FORMULA: TWO-PROPORTION SQUARED Z-STATISTIC TO PEARSON CHI-SQUARE (2/4)

$$z^2 = \left(\frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}} \right)^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \chi^2$$

Symbols:

χ^2 : Pearson chi-square test statistic with $(r - 1)(c - 1) = 1$ degree of freedom

Inputs: CedarCart conversion counts: $n_A = 400$, $x_A = 160$, $n_B = 400$, $x_B = 200$, pooled proportion $p_0 = 360 / 800 = 0.45$.

Steps:

1. Compute pooled proportion: $p_0 = 360 / 800 = 0.4500$.
2. Calculate null standard error: $SE_0 = \sqrt{0.45 * 0.55 * (1/400 + 1/400)} = \sqrt{0.2475 * 0.005} = \sqrt{0.0012375} = 0.035178$.

Not in your PNP: follow along; no written response here.

EQUIVALENCE FORMULA: TWO-PROPORTION SQUARED Z-STATISTIC TO PEARSON CHI-SQUARE (3/4)

$$z^2 = \left(\frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}} \right)^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \chi^2$$

Steps:

3. Compute z-score: $z = (0.5000 - 0.4000) / 0.035178 = 0.1000 / 0.035178 = 2.842676$.

4. Square the z-score: $z^2 = (2.842676)^2 = 8.080808$.

5. Calculate expected counts for 2x2 table: $E_{11} = (400 * 360) / 800 = 180$, $E_{12} = 220$, $E_{21} = 180$, $E_{22} = 220$.

6. Sum squared deviations: $(160-180)^2 / 180 + (240-220)^2 / 220 + (200-180)^2 / 180 + (200-220)^2 / 220 = 400/180 + 400/220 + 400/180 + 400/220 = 2.2222 + 1.8182 + 2.2222 + 1.8182 = 8.080808$.

Not in your PNP: follow along; no written response here.

EQUIVALENCE FORMULA: TWO-PROPORTION SQUARED Z-STATISTIC TO PEARSON CHI-SQUARE (4/4)

$$z^2 = \left(\frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}_0(1 - \hat{p}_0) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}} \right)^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \chi^2$$

Steps:

7. Confirm equality: $z^2 = 8.080808 = \chi^2$.

Result: $z^2 = 8.081 = \chi^2$ with $df = 1$ and $p = 0.0045$.

The squared two-proportion z-statistic equals the Pearson chi-square statistic exactly, demonstrating that Excel z-test formulas and SPSS cross-tabulation routines execute identical mathematics.

Use: Use this algebraic identity to verify that spreadsheet models and statistical software packages are perfectly reconciled before presenting findings to leadership.

Limit: This exact equality holds strictly for the uncorrected Pearson chi-square statistic; applying Yates continuity correction breaks the algebraic equivalence.

Not in your PNP: follow along; no written response here.

AI SEARCH & SHARE



Discuss with a partner;
artificial intelligence (AI)
support is optional.

APPLY THE IDEA

Discuss with a partner; artificial intelligence (AI) support is optional.

1

Key point

- Why did CedarCart management reject Variant B even though both Excel and **SPSS** confirmed that its purchase uplift was statistically significant with $p = 0.0045$?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

Not in your PNP: follow the activity directions here.

DECISION POINT: PRACTICAL SIGNIFICANCE, GUARDRAIL METRICS, AND CROSS-TOOL RECONCILIATION

Use the evidence, explain your decision, and name one limit.



Key Point

Variant B lifted conversion significantly ($p = 0.0045$), but complaints surged from 2% to 8% (guardrail limit: $\leq 1\%$ increase). Should CedarCart launch Variant B?

In your PNP: use the matching slide page.



Excel first, SPSS second

LAB PRACTICE 3: GUARDRAIL AUDIT, NET IMPACT, AND TOOL RECONCILIATION

10 minutes in the lab; then slide 78.

1

Open and do

- **Colab:** heading Class practice 3: check complaints and net impact. Excel cells: **B7, B8, C7, D7**.
- **Excel:** Calculate complaint proportions in **B9:B10**, net profit impact in **B13**, and record decision recommendation. **SPSS:** Run **CROSSTABS** for group by complaint with **CHISQ** in Block.

2

Check and bring back

- Calculated net loss of -\$360 and verified guardrail breach leading to recommendation against rollout. Timer 10 min, debrief 2 min, then return to the group studio (slide 78).

3

Help or go further

- **If you need help:** Open Toolkit Section 4.1, review the net economic formula, and calculate complaint costs for 24 extra tickets.
- **If you finish early:** Critique this decision: how does an edge condition of doubled complaint costs alter the net economic test outcome?

In your PNP: use this page for the lab handoff.



PRACTICE

Sheet: IN-CLASS_Lab Part C

Not in your PNP: follow the activity directions here.

LAB PART C: AUDIT THE LAUNCH DECISION | CONTINUED 2

Sheet: IN-CLASS_Lab Part C



Sheet: IN-CLASS_Lab Part
C

1

Expected checks

- Complaints rise by 6.0 percentage points, net impact is $-\$360$ per 400 B users, and the squared Excel z-statistic rounds to the **SPSS** value of 8.081.
- Stop at the lab guide's Part C stop point; the group studio follows.

Not in your PNP: follow along; no written response here.



GROUP STUDIO

Use your issued worksheet and assigned group

Not in PNP-IND: use the PNP-GRP document.



Purchase gains, complaint
harm and the launch
decision

GROUP STUDIO: SHOULD CEDARCART LAUNCH B?

12 minutes group work; 5 minutes whole-class sharing

1

Roles and file

- Facilitator, Evidence checker, Recorder, Reporter. Open the **PNP-GRP** practice document.

2

Use today's lab

- Verify that conversion difference is stated in percentage points, not relative percent. Check whether the lower bound of the conversion confidence interval meets the 8.0 point hurdle. Evaluate whether customer support complaint costs exceed incremental gross margin gains.

3

Bring back

- A concise executive decision on checkout Variant B specifying the statistical conversion evidence, the complaint guardrail breach, and the net economic loss. Then synthesis (slide 80) and the exit ticket in your **PNP-IND** guided notes.

Not in PNP-IND: use the PNP-GRP document.



Save, resume, reflect

CONTINUE FROM YOUR SAVED CHECKPOINT

The Take home portion completes this session's lab

1

Save before leaving

- Use the checkpoint named in your lab's In class route.

2

Resume at home

- **task_04:** CedarCart Practice Revisit and Re-estimation
- IN-CLASS_Lab Part C!**B15**.

3

Check and submit

- Complete the listed checks and follow the issued submission instructions. Use the catch-up route if needed.

Not in your PNP: follow along; no written response here.

REVISIT THE CEDARCART DECISION

Final vote and evidence synthesis

- **A: Deploy B now**

Benefit: B produced more purchases in the test. Risk: The complaint ceiling was breached and stated net impact was negative.

- **B: Hold B for revision**

Benefit: The team can investigate complaints before another test. Risk: It delays a possible purchase gain.

- **Your final vote**

Would you deploy variant B nationwide now or hold it for revision, and which evidence should decide?

Not in your PNP: follow along; no written response here.

REVISIT THE CEDARCART DECISION | CONTINUED 2

Final vote and evidence synthesis



Final vote and evidence
synthesis

1

Your final vote

- The benchmark test did not verify the satisfaction goal. The purchase test detected a 10.0-percentage-point gain, but its interval extends below the 8-point practical hurdle.
- Complaints rose 6.0 points against a 1.0-point ceiling, and net impact was –\$360 per 400 B users. The recommendation is bounded to the tested app population and stated assumptions.

Not in your PNP: follow along; no written response here.



One optional question
remains

THANKS, AND LOOKING AHEAD

One optional question remains

1

Key point

- Today's conclusion is to hold CedarCart's current variant B for revision, not to treat a significant p-value as launch approval.

2

Optional question, not homework

- What evidence from a revised checkout test would justify reconsidering B for the same eligible mobile app population?

Not in your PNP: follow along; no written response here.



Reference slide

SESSION MATERIALS

Reference slide

1

Key point

- **GC1015** Session 6 distributed slides and named student companions.
- Issued Excel workbook and lab guide, including **IN-CLASS_Lab Part C**; issued group-studio worksheet and rubric.

2

Key point

- Excel 365 and **IBM SPSS** Statistics procedures used in this session.

Not in your PNP: follow along; no written response here.

APPENDIX

APPENDIX A: SELECTED STUDY SUMMARIES

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.

Key point

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.

Not in your PNP: use the Session Study Toolkit after class.



Formulating Hypotheses
and Error Trade-offs

CEDARCART WORKED DECISION SPECIFICATION

Formulating Hypotheses and Error Trade-offs

1

Prespecifying Hypotheses and Alpha

- **CedarCart defines H_0 :** p_B minus p_A equals 0 against **H_1 :** p_B minus p_A does not equal 0 at α equals 0.05 prior to data collection.

2

Practical Uplift Hurdle

- Beyond statistical significance, CedarCart requires an estimated uplift of at least 8 percentage points to warrant rollout overhead.

3

Operational Harm Guardrail

- Customer complaints must not rise by more than 1 percentage point. All four criteria must align before variant B can launch.

Not in your PNP: use the Session Study Toolkit after class.



Randomized Units and
Two-Proportion z-Test
Mechanics

COMMON MISCONCEPTION: PERCENTAGE POINTS VERSUS PERCENTAGE CHANGE

Randomized Units and Two-Proportion z-Test Mechanics

1

Avoid the Metric Trap

- Never confuse percentage points with relative percent change. Absolute difference: $50\% - 40\% = 10$ percentage points. Relative uplift: $(50\% - 40\%) / 40\% = 25\%$ relative gain.

2

Executive Clarity

- Reporting 10% when you mean 10 percentage points severely understates performance; reporting 25 percentage points when you mean 25% relative uplift drastically inflates unit volume.

Not in your PNP: use the Session Study Toolkit after class.



Interval Estimation for
Absolute Difference

COMMON FALLACIES IN UPLIFT REPORTING

Interval Estimation for Absolute Difference

1

Error 1: Standard Error Confusion

- Do not reuse the pooled standard error ($SE_0 = 0.03518$) to build confidence intervals.
- Intervals must reflect distinct group variances using the unpooled formula ($SE_{CI} = 0.0350$).

2

Error 2: Points Versus Percentages

- CedarCart achieved a 10 percentage point uplift, which equals a 25 percent relative increase.
- Calling this a '10% lift' understates relative performance and creates severe financial forecasting errors.

Not in your PNP: use the Session Study Toolkit after class.



Practical Significance and
A/B Guardrails

COMMON MISCONCEPTIONS IN EXPERIMENT EVALUATION

Practical Significance and A/B Guardrails

1

Misconception 1: P-Value Equals Commercial Launch

- A statistically significant difference only filters out random noise. It does not prove that an effect covers operational expenses or creates enterprise value.

2

Misconception 2: Ignoring Guardrail Side Effects

- Focusing solely on conversion rates ignores downstream consequences, such as support ticket spikes, refunds, or customer churn.

3

Misconception 3: Treating Missing Data as Zeroes

- Blank records must never be silently converted into zeroes. A missing checkout record may represent a software crash that distorts analysis.

Not in your PNP: use the Session Study Toolkit after class.



Tool Reconciliation and
Reproducibility in Excel
and SPSS

COMMON DATA PIPELINE PITFALLS: SILENT BLANKS AND RATIO MISMATCH

Tool Reconciliation and Reproducibility in Excel and SPSS

1

Missing Data Handling

- Excel **COUNTIFS** treats blank text strings differently than numeric zeroes. Converting missing records to zeroes silently corrupts the sample denominator and skews conversion rates.

2

Software Filtering Discrepancies

- **SPSS** excludes missing values by default, shrinking the effective sample size without warning if data pipelines drop records. Always verify total case counts against the assignment log.

3

Sample-Ratio Mismatch (SRM)

- A significant deviation from the intended 50-50 allocation indicates a broken pipeline or selective user drop-off. A valid test cannot proceed on data with an unresolved **SRM**.

Not in your PNP: use the Session Study Toolkit after class.

APPENDIX

APPENDIX B: EXPLORATION AND STRETCH

Optional independent study

Key Point

Optional selected readings: Check the Traffic Split; A Split That Merits Attention

Not in your PNP: use the Session Study Toolkit after class.

CHECK THE TRAFFIC SPLIT (1/2)

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}, \quad E_i = Nr_i, \quad df = k - 1$$

Symbols:

O_i: observed randomized users in arm i; **E_i**: expected users.

N: total users; **r_i**: planned share; **k**: number of arms; **df**: degrees of freedom.

Inputs: Planned arm shares and observed counts of unique randomized users.

Steps:

1. Compute expected counts from the planned shares.

CHECK THE TRAFFIC SPLIT (2/2)

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}, \quad E_i = Nr_i, \quad df = k - 1$$

Steps:

2. Sum each arm's squared observed-minus-expected difference divided by its expected count.

Result: Compare the statistic with a chi-square distribution with $k-1$ degrees of freedom.

An unusually small p-value signals an allocation anomaly to investigate.

Use: Screen experiment traffic before evaluating conversion uplift.

Limit: The check does not diagnose the cause or establish that outcomes are unbiased.

Not in your PNP: use the Session Study Toolkit after class.

A SPLIT THAT MERITS ATTENTION

Hypothetical 50/50 allocation check

Arm	Observed users	Expected users	Chi-square term
A	400	430	2.093
B	460	430	2.093

Each arm differs from its planned count by 30 users; the combined chi-square statistic is 4.186.

Unique randomized users in two hypothetical experiment arms | Teaching illustration | Unit: users

Not in your PNP: use the Session Study Toolkit after class.



Hypothetical 50/50
allocation check

A SPLIT THAT MERITS ATTENTION

Hypothetical 50/50 allocation check

1

Inputs

- The assignment log contains 400 unique users in A and 460 in B, or 860 total. A planned 50/50 split expects 430 in each arm.

2

Calculation

- $\chi^2 = 900/430 + 900/430 = 4.186$. With one degree of freedom, $p \approx 0.041$. That is below 0.05, but above the conservative 0.0005 screening threshold described in Microsoft's guidance.

3

Interpretation

- This threshold would not flag the split for an automatic stop. It does not certify the data: check assignment logs and denominators if other integrity concerns appear.

Not in your PNP: use the Session Study Toolkit after class.

SECTION

INDUSTRY APPLICATION

Optional post-session reading - outside live class time

Not in your PNP: use the Session Study Toolkit after class.



WHEN SEARCH SLOWS

Reported evidence from Bing

1

Situation.

- Microsoft researchers used controlled experiments on production Bing search traffic to study what happens when search responses are deliberately slowed.
- Their 2013 paper reports effects on user behavior and business outcomes, making response speed relevant to product decisions rather than merely an engineering concern.

2

Measure and interpretation.

- Latency is response delay, measured in milliseconds (ms). The case shows why a feature's primary gain should be considered alongside its effect on speed.

3

Limit.

- The cited paper describes experiments reported in 2013; an exact experiment calendar window is not specified here. It supplies neither a universal launch ceiling nor a current revenue forecast.
- **Source:** [Microsoft researchers, 2013](<https://doi.org/10.1145/2487575.2488217>).

Not in your PNP: use the Session Study Toolkit after class.



APPLY A SPEED GUARDRAIL

Illustrative rollout decision, not Bing results

1

Illustrative test.

- Suppose a search team tests a new ranking feature among eligible United States search-result impressions during one hypothetical week. Click-through rate means clicks divided by impressions.
- **It rises from 5.00% to 5.06%:** $5.06\% - 5.00\% = 0.06$ percentage points, or $0.06 / 5.00 = 1.2\%$ relative uplift.

2

Guardrail decision.

- Suppose the team set a maximum additional response delay of 150 milliseconds before testing. The feature adds 180 milliseconds: $180 - 150 = 30$ milliseconds beyond the limit.
- Do not roll it out on these inputs, regardless of the click gain.

3

Limit.

- These figures are invented for instruction; no statistical test or revenue effect is established.

Not in your PNP: use the Session Study Toolkit after class.